Ram Spectral Certificates for a Smooth Banach Construction: M279

Exact Haar Correlations, Density-Zero Whitening,
Hellinger Refinements, and Moving-Window Cyclicity

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M279

October 7, 2026

Abstract

The public manuscript M279 claims a smooth volume-preserving diffeomorphism of the three-torus with simple Lebesgue spectrum on the entire centered L^2 space. This is exactly the probability-preserving Banach problem, not merely the construction of an absolutely continuous spectral component. The local Ram report TR-item-169 supplies directly applicable correlation, cyclicity, and physical frame criteria. We trace the pinned manuscript's complete mathematical source, including its stationary-phase passages, positive Fourier signal law, centered prediction blocks, zero-set repair, and retention of previously chosen Laurent predictions.

Several restricted results are proved independently here. The optimal global quadratic minorant of the square root halves one source loss estimate. Retaining the actual color-weighted second moment improves the ideal block Hellinger loss to $3\delta^2$, and the accumulated actual loss to $4K\delta^2$ under explicit per-block realization budgets. The intermediate $150\delta^2$ and $26\delta^2$ estimates explain the separate improvements. Spectral whitening with zero-set repair is characterized as a nearest flat-vector construction. A Gram-normalized moving-window certificate, exact residual formulas, a targeted prediction-retention theorem, and a counterexample to an insufficient strong-limit argument complete the physical error account.

These conclusions are not an independent certification of the manuscript's global existence theorem. The elementary refinements have proofs below; their implementation in the claimed smooth construction remains conditional on the passage and signal realization estimates and their parameter choices. Full-source reading, finite exact checks, established external results, and unverified analytic dependencies are distinguished throughout. The Ram formulation uses classical spectral, Fourier, frame, and matrix theory; no new dynamical realization follows from a change of transform coordinates.

Contents

1 The exact target and the status of the conclusions **3**
- 1.1 Three levels of assertion . . . 3
- 1.2 Public literature and the relevant distinctions . . . 4

2 Exact local Ram grounding **4**
- 2.1 What "Ram rederivation" means in this problem . . . 5

3 The source passage and signal chain **5**
- 3.1 Packets, domains, and the exact spectral identity . . . 5
- 3.2 A genuine volume-preserving passage . . . 6
- 3.3 Fast averaging is not pointwise control . . . 7
- 3.4 The prime-sieve Fourier signal . . . 7

4 Centered blocks and a sharper proved Hellinger budget **8**
- 4.1 The ideal finite-phase experiment . . . 8
- 4.2 Unit-disk centering is already available . . . 9
- 4.3 Actual frequencies and actual error, not independent randomness . . . 10
- 4.4 Prediction variance and the accumulated improvement . . . 11

5 Whitening at density zero: existence, distance, and stability **12**
- 5.1 A flat vector already exists for each intermediate twist . . . 12
- 5.2 Why the zero repair cannot be omitted . . . 13
- 5.3 Laurent prediction after whitening . . . 14

6 Physical Gram certificates for moving orbit windows **14**
- 6.1 Exact synthesis and residual formulas . . . 14
- 6.2 A directional certificate for a chosen predictor . . . 16
- 6.3 Why a finite Ram panel cannot certify the whole orbit space . . . 16

7 Passing to the smooth limit without losing cyclicity **16**
- 7.1 An explicit counterexample to fixed-power reasoning alone . . . 16
- 7.2 The targeted criterion actually used by M279 . . . 17
- 7.3 Smooth physical convergence and the source's explicit tail . . . 18

8 What the one-stage construction still has to supply **19**

9 Consequences with their exact hypotheses **20**

10 Evidence, limits, and practical next steps **20**
- 10.1 Reading and verification depth . . . 20
- 10.2 Advantages and costs of the concrete extensions . . . 22
- 10.3 A focused future research program . . . 22
- 10.4 Bottom line . . . 23

A Source manifest and exact locators **23**
- A.1 Pinned public manuscript and actual reading . . . 23
- A.2 Confidential local corpus: actual files, not just maps . . . 24
- A.3 Public-reference access depth . . . 25

1 The exact target and the status of the conclusions

Let $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, let μ be normalized standard volume on $\mathbb{T}^3$, and put

$$\mathcal{H} = L^2_0(\mathbb{T}^3, \mu; \mathbb{C}) = \{g \in L^2 : \int g \, d\mu = 0\}.$$

For an invertible volume-preserving map T, its Koopman operator is $U_T g = g \circ T$. It is unitary on $\mathcal{H}$. The source's `01-introduction.tex`, Theorem `thm:main`, asserts the existence of a C^∞ volume-preserving diffeomorphism T and a *real* $f \in \mathcal{H}$ such that

$$\{U_T^n f : n \in \mathbb{Z}\} \quad \text{is an orthonormal basis of } \mathcal{H}. \tag{1.1}$$

The observable is only required to belong to L^2; it is not asserted to be smooth. Constants have been removed. This distinction is indispensable: constants are invariant eigenvectors and cannot belong to a bilateral Lebesgue-shift basis.

We use inner products linear in the first argument. If E_U is the spectral resolution of a unitary U, write $\sigma_g(B) = \langle E_U(B)g, g\rangle$. On the temporal circle $\mathbb{T}$, with normalized Haar measure m, our convention is

$$\langle U^n g, g\rangle = \int_{\mathbb{T}} e^{2\pi i n z} \, d\sigma_g(z). \tag{1.2}$$

The two distinct requirements in (1.1) are

$$\sigma_f = m, \qquad \overline{\operatorname{span}}\{U^n f : n \in \mathbb{Z}\} = \mathcal{H}. \tag{1.3}$$

The first gives one Haar component; the second says there is no remaining orthogonal spectral component. A finite matrix, an abstract shift, a flow, or a sigma-finite measure construction is not automatically a solution of this specified problem.

1.1 Three levels of assertion

1. **Source claims.** The geometric existence theorem and the entire passage–signal–prediction construction are claims of the pinned public manuscript [1]. We reconstruct their logical role and examine the specified vulnerable steps, without reporting a complete analytic validation.
2. **Results proved here.** The theorems explicitly proved in this report are Hilbert-space, finite-moment, and scalar inequalities under their stated hypotheses. They do not assume that a proposed smooth construction has already supplied those hypotheses.
3. **Conditional consequences.** Applying the sharper block bounds to an actual passage sequence, or applying the completion criterion to an actual smooth limit, requires the source realization estimates. Ordinary mixing follows from (1.1); entropy and Lyapunov assertions use additional classical theorems. Mixing of all orders requires a separate argument not verified in this report.

The catalogue's level 5 applicability is justified: TR-item-169 concerns the same probability-space Koopman problem and supplies a concrete physical moving-window criterion. Applicability is neither a validity verdict nor a claim that the Ram corpus already supplies the missing construction.

1.2 Public literature and the relevant distinctions

The literature check was made against primary public records on October 7, 2026, using public titles and identifiers only. Ryzhikov's 2024 survey, Section 1, Q2, explicitly distinguishes the infinite-measure Banach–Ulam formulation from the probability-space question on the orthogonal complement of constants [6]. That is the precise distinction used here, not an inference from the phrase "Lebesgue spectrum."

Prikhod'ko's published 2020 result concerns probability-preserving *flows* with simple Lebesgue spectrum [7]. A nonzero time map of the model $M_{e^{2\pi i t\xi}}$ on $L^2(\mathbb{R}, d\xi)$ has countably infinite Lebesgue multiplicity: the substitution $\xi = (z+k)/|t|$, $z \in [0,1)$, $k \in \mathbb{Z}$, splits the space into countably many circle copies (reversing orientation when $t < 0$). Thus the flow theorem does not imply M279.

Fayad–Forni–Kanigowski construct a real analytic conservative flow on $\mathbb{T}^2$ with one singularity and countably infinite Lebesgue multiplicity; their Theorem 1 and the multiplicity criterion in Section 7 distinguish type from multiplicity [8]. Abdedou–Fayad–Kessi obtain a real analytic reparametrization of a minimal translation flow on $\mathbb{T}^5$, also of infinite multiplicity [9]. Abdedou–Fayad–Thouvenot's 2026 English publication shows that every ergodic system induces a system of pure infinite-multiplicity Lebesgue spectrum [10]. None is a multiplicity-one discrete-time realization on the full centered space.

The preprint *Spectral ergodic Banach problem and flat polynomials* advertises an infinite conservative measure result [11]; its global proof was not independently assessed here. The same caution applies to replacing M279's pinned claim by a statement that the probability problem is now generally accepted as settled. This report establishes neither historical priority nor community acceptance. Public records and selected statements were checked; these external papers were not all read in full.

2 Exact local Ram grounding

The source manifest in Appendix A gives absolute paths and exact labels. The decisive local source is [2], read in full. Its Section 3, *Equivalent correlation criterion*, identifies (1.3). Section 14, *Moving-window cyclicity with a physical frame budget*, corrects the insufficient earlier finite-to-limit criterion by adding a physical synthesis-transfer estimate. Sections 11–15 separate probability preservation, continuous-time results, spectral type, cyclicity, and finite numerical evidence.

Two additional foundation sources were read at the actual theorem level, not inferred from maps. The revised TR248's `thm:parseval`, `eq:leak`, `thm:finite`, and `thm:perturb` give complete-bank Parseval identities, finite-model leakage, and perturbation margins [3]. Ram Calculus Volume 2, Chapter 2, `v2:c02:frame-operators`, `v2:c02:canonical-decoder`, and `v2:c02:coefficient-projector` distinguish analysis, synthesis, dual inversion, and projection onto consistent coefficients [4]. Volume 21's physical-metric chapter supplies `v21c01:physical-gram-definition` and `v21c01:residual-identity`: the Gram metric and the omitted physical residual must remain in a finite representation [5].

2.1 What "Ram rederivation" means in this problem

On a bounded chart patch, Ram exponential-polynomial atoms have the form $w(x)e^{s\cdot(x-c)}p(x-c)$. With real bounded w, finite physical measure, and $g \in L^2$, define the bilinear moment generator

$$\mathcal{R}_{g,w}(z) = \int g(x)w(x)e^{z\cdot(x-c)}\,d\mu(x).$$

If $p(t) = \sum_\alpha d_\alpha t^\alpha$, the Hermitian measurement is exactly

$$\langle g, we^{s\cdot(x-c)}p(x-c)\rangle = \sum_\alpha \overline{d_\alpha}\,\partial_z^\alpha \mathcal{R}_{g,w}(z)\Big|_{z=\bar{s}}. \tag{2.1}$$

Bounded support and finite measure justify each finite derivative by dominated convergence. This is the revised TR248's atom convention and its moment-Gram appendix, not a new integral identity.

For M279 the useful specialization is the angular Fourier slice and its physical orbit synthesis. Given $\Lambda \subset \mathbb{Z}$ finite, the columns are $U_T^n f$, $n \in \Lambda$, not arbitrary algebraic monomials. Their exact Gram matrix uses the correlations (1.2); the whole question is whether the resulting physical spans approximate all of $\mathcal{H}$ and survive the changing-map limit. Relabeling Fourier coefficients as Ram coefficients does not make them cyclic. Likewise the general moment formula does not evaluate a stationary-phase integral exactly, establish a frame bound on all of L^2, or prove a smooth measure-preserving realization.

The additional value of the local framework is specific: it exposes the physical norm, mass matrix, retained subspace, model remainder, and transfer error. The underlying Fourier, spectral, and frame ingredients are classical. No nonlinear integral-equation or analytic-continuation theorem is needed merely because those topics occur elsewhere in the corpus.

3 The source passage and signal chain

This section records the mathematical mechanism actually used by M279, with the exact hypotheses needed later. It is a reconstruction of source arguments, not an independent proof of every uniform asymptotic estimate.

3.1 Packets, domains, and the exact spectral identity

An intermediate map, in a volume-preserving chart, is

$$S(\theta,u,a) = (\theta+\phi(u),u,a), \qquad F(\theta,u,a) = A(u,a)e^{2\pi i m\theta}, \quad m \in \mathbb{Z}\setminus\{0\}. \tag{3.1}$$

Here ϕ is smooth and real on the circle, $\phi' > 0$ on a good arc I, and A is supported compactly in $I\times\mathbb{T}$. For the smooth passage estimates A is smooth; the elementary spectral formula only requires A in L^2. Direct integration in θ gives

$$\sigma_F = (m\phi \bmod 1)_*(|A(u,a)|^2\,du\,da), \qquad v(U_S)F = v(m\phi(u))A(u,a)e^{2\pi i m\theta}. \tag{3.2}$$

The second formula holds on the functional-calculus domain $\int |v|^2\,d\sigma_F < \infty$. In particular, an unbounded reciprocal square root is not excluded, but its domain must be checked.

Different angular indices are orthogonal, as are equal-index packets with disjoint coefficient supports. The same statements hold after any spectral multiplier, so the corresponding cyclic subspaces are orthogonal. This is why color densities add and why tagged prediction error has the exact form

$$\|\sum_r b_r F_r - \Psi(U_S)\sum_r F_r\|^2 = \sum_r \int_{\mathbb{T}} |b_r - \Psi(z)|^2 p_r(z)\,dz. \tag{3.3}$$

The notation on the left means the squared Hilbert norm of the displayed vector. These statements correspond to source `eq:packet-orthogonality` and `eq:packet-spectral-measure`. For different lists whose supports overlap, only within-list orthogonality is available; the source explicitly allows simultaneous transport without asserting cross-list orthogonality.

For the *same* unitary U, the spectral-measure total variation obeys

$$\|\sigma_F - \sigma_G\|_{\mathrm{TV}} \le (\|F\| + \|G\|)\|F - G\|.$$

Indeed test against a bounded spectral multiplier of norm at most one, subtract the two quadratic forms, and split into terms containing $F - G$. Here total variation is the L^1 difference when densities exist, without the probability-distance factor $1/2$. This rederives source `eq:density-norm-bound`. It does not compare spectral densities of two different nearby maps; that task requires the separate passage estimates.

3.2 A genuine volume-preserving passage

Choose an integer c separating the finitely many ranges $m(c + \phi(\mathbb{T}))$ and keeping them away from zero. Set $V = c + \phi$. For a large integer N, choose $|\Delta_N| \le N^{-1}$ such that $N(\int V + \Delta_N) \in \mathbb{Z}$, and choose a lift H_N with $H_N' = N(V + \Delta_N)$. With a fixed smooth jitter J, put

$$\sigma_N(a) = N^{-1} + N^{-2}J(a), \tag{3.4}$$

$$T_N(\theta, u, a) = (\theta + H_N(u + \sigma_N(a)) - H_N(u), u + \sigma_N(a), a), \tag{3.5}$$

$$C_N(\theta, u, a) = (u, a, \theta - H_N(u)). \tag{3.6}$$

The integral winding condition makes C_N well-defined on the torus. It has Jacobian one and conjugates T_N to a twist of speed σ_N. For the fixed chart and fixed data, T_N and T_N^{-1} converge in every C^k norm to S and S^{-1}. The integer c disappears modulo one. Future charts can have large derivatives; the source does not claim a chart-uniform convergence rate independent of the preceding construction.

The exact transformed coefficient is

$$C_{mj}(a) = \int_{\mathbb{T}} A(u, a) e^{2\pi i m H_N(u) - ju}\,du. \tag{3.7}$$

Stationary phase applies on the good arc to $m(V(u) + \Delta_N) = j/N$, with nonzero second derivative $m\phi'(u)$. The leading squared mass is $N^{-1}W(j/N - m\Delta_N, a)$, where

$$W(s, a) = \frac{|A(u, a)|^2}{|m\phi'(u)|}, \qquad s = mV(u),$$

extended by zero. Source `lem:passage` proves a summed L^1 mass comparison, including the nonstationary tails and compact cutoffs. A pointwise leading term alone would not suffice: there are $O(N)$ retained indices, and the relevant error must still vanish after summation.

3.3 Fast averaging is not pointwise control

Let ψ be a smooth cutoff on an arc where $J' > 0$. Pushing $W(s,a)\psi(a)^2\,da$ by $a \mapsto sJ(a)$ gives a compactly supported continuous density $K(s,y)$ with Jacobian $|s|J'(a)$. The leading unfolded temporal density in a residue class $j \equiv h \pmod P$ is

$$Q_{N,h}(t) = \sum_{k\in\mathbb{Z}} K(t, Nt - h - kP). \tag{3.8}$$

Only a bounded number of terms contribute at each t on the relevant compact range. Source `eq:rounding-mass-removal` removes $m\Delta_N$ in the coefficient-mass array *before* sampling the fast variable. Replacing a fast phase by a nearby slow-coordinate value without this step would not follow from $\Delta_N = O(N^{-1})$.

Source `lem:fast-average` applies to a continuous, compactly supported slow-variable function periodic in the fast variable. It identifies the limit of the integral along $y = Nt + \beta_N$ with the period average, uniformly in offsets β_N. Consequently the L^1 error from replacing a real-line jitter law by a smooth compact realization is controlled *after diagonal averaging*; small total variation of laws is not a uniform pointwise bound for every sampled periodization. In the neutral passage the jitter width is fixed large first and N is chosen last. The residue densities can then approach p/P while physical perturbation and the new bad arc are as small as prescribed.

3.4 The prime-sieve Fourier signal

For a fixed normalized frequency band $b \le |m|/B \le D_0$ and a smooth compactly supported d away from zero, with $d(-x) = \overline{d(x)}$, source `lem:signal` constructs a positive law. The baseline q_0 is a strictly positive probability density with quadratic tails and Fourier transform supported in $[-\alpha,\alpha]$, $\alpha < b/2$. Let g be the real inverse Fourier transform of d and $C_P(h)$ the real Ramanujan sum. For a sufficiently small $\gamma > 0$,

$$Q_h(R) = \frac{q_0(R) + \gamma C_P(h) g(R)}{P} > 0, \quad \int Q_h = \frac{1}{P}. \tag{3.9}$$

Here P is the product of the primes up to an integer L larger than the ratio of support bandwidth to b. Orthogonality of characters on $\mathbb{Z}/P\mathbb{Z}$ gives the exact identity

$$\sum_h \int Q_h(R) e^{2\pi i v R + kh/P}\,dR = \widehat{q}_0(v)\mathbf{1}_{P|k} + \gamma d(v)\mathbf{1}_{\gcd(k,P)=1}. \tag{3.10}$$

The Fourier sign is the source's positive-exponent convention in this formula. All higher potentially active integer harmonics have a prime divisor in P, so the ideal nonconstant law retains only the desired first harmonic. With c finite rather than infinite, a finite-band error must be controlled before realizing the law with smooth J and ψ.

After a neutral residue refresh the useful amplitude is $\lambda = \gamma\varphi(P)/P$. The signal strength can be fixed independently of the later large angular scale B, but the geometry may depend on the actual finite set of indices. The order is important: fix band data and strength, then actual-index separations and drift tolerances, then jitter, and finally N. The source's temporal-drift bound contains a support enlargement term *not* divided by N. Increasing N alone cannot repair an overly large support cutoff.

The required source locators are `eq:signal-joint-fourier`, `eq:signal-exact-moment`, `eq:signal-finite-c`, `eq:signal-index-drift`, `eq:signal-temporal-drift`, and `lem:refresh-`

`transmit`. Reading these calculations identifies a plausible and explicit route; it does not replace an independent proof of every uniform stationary-phase, law-approximation, and support-separation estimate.

4 Centered blocks and a sharper proved Hellinger budget

We now separate the elementary finite-phase calculation from its geometric realization. This permits a concrete improvement without claiming a new solution of the global problem.

4.1 The ideal finite-phase experiment

At a fixed temporal point z, let $t_r \geq 0$, $\sum_r t_r = 1$, and let $|b_r| \leq 1$. In the application $t_r = p_r/p$ where $p = \sum_r p_r$; values on $\{p = 0\}$ can be set arbitrarily. Put

$$\beta = \sum_r t_r b_r, \quad a_r = b_r - q, \quad H_r = \prod_{i=1}^{n} [1 + \lambda(a_r \bar{Z}_i + \bar{a}_r Z_i)], \quad R = \sum_r t_r H_r,$$

where the Z_i are independent uniform circle variables. Suppose $|a_r| \leq A$, $2\lambda A < 1$, and

$$0 < \delta \leq \frac{1}{100}, \qquad \lambda^2 \leq \frac{\delta}{2}, \qquad n = \left\lfloor \frac{\delta}{\lambda^2} \right\rfloor. \tag{4.1}$$

Then $\delta/2 \leq n\lambda^2 \leq \delta$. The factors are positive and

$$\mathbb{E}H_r = 1, \quad \mathbb{E}R = 1, \quad \mathbb{E}(H_r H_s) = [1 + 2\lambda^2 \operatorname{Re}(a_r \bar{a}_s)]^n. \tag{4.2}$$

These identities use only integration of characters of powers $0, \pm 1, \pm 2$, not a probabilistic assertion about the actual deterministic frequencies.

Lemma 4.1 (Variance bound with an explicit amplitude parameter)**.** *Under* (4.1) *and the preceding assumptions,*

$$\mathbb{E}(R-1)^2 \leq 2\delta|\beta - q|^2 + 2A^4 e^{2A^2\delta}\delta^2. \tag{4.3}$$

More sharply, if $S = \sum_r t_r |a_r|^2$*, then*

$$\mathbb{E}(R-1)^2 \leq 2\delta|\beta - q|^2 + 2e^{2A^2\delta}\delta^2 S^2. \tag{4.4}$$

Proof. For real y with $|y| < 1$ and integer $n \geq 1$, Taylor's formula (with the case $n = 1$ exact) yields

$$|(1+y)^n - 1 - ny| \leq \frac{n(n-1)}{2} y^2 (1+|y|)^{n-2} \leq \frac{n^2 y^2}{2} e^{n|y|}.$$

The first inequality for $n \geq 2$ follows by bounding the second derivative on the segment joining zero to y. Here $y_{rs} = 2\lambda^2 \operatorname{Re}(a_r \bar{a}_s)$ satisfies $|y_{rs}| \leq 2A^2\lambda^2 < 1$ and $n|y_{rs}| \leq 2A^2\delta$. The weighted linear term is $2n\lambda^2 |\sum_r t_r a_r|^2 = 2n\lambda^2|\beta - q|^2$. For the sharper estimate, retain $y_{rs}^2 \leq 4\lambda^4 |a_r|^2 |a_s|^2$ instead of replacing every amplitude by A. The weighted sum of Taylor remainders is then at most $2\delta^2 e^{2A^2\delta}(\sum_r t_r |a_r|^2)^2$. Use (4.2) and $\mathbb{E}R = 1$ to obtain (4.4); the weaker bound (4.3) follows from $S \leq A^2$. □

Lemma 4.2 (Sharp global quadratic square-root minorant)**.** *For all $x \geq 0$,*

$$\sqrt{x} \geq 1 + \frac{x-1}{2} - \frac{(x-1)^2}{2}. \tag{4.5}$$

The coefficient $1/2$ is the smallest constant valid globally in this form. More generally the smallest coefficient valid for all $x \geq m \geq 0$ is $[2(1+\sqrt{m})^2]^{-1}$.

Proof. Writing $t = \sqrt{x}$, the difference between the left side of (4.5) and its right side is $t(t-1)^2(t+2)/2 \geq 0$. At $x = 0$ the proposed general coefficient c would require $0 \geq 1/2 - c$, so $c \geq 1/2$. For the floor statement, use the exact identity

$$1 + \frac{x-1}{2} - \sqrt{x} = \frac{(x-1)^2}{2(1+\sqrt{x})^2}.$$

The quotient is decreasing in x. At $m = 1$ its sharpness follows by taking $x \downarrow 1$; otherwise evaluation at $x = m$ proves sharpness directly. □

If $R \geq 0$ and $\mathbb{E}R = 1$, the lemma gives

$$1 - \mathbb{E}\sqrt{R} \leq \frac{1}{2}\mathbb{E}(R-1)^2. \tag{4.6}$$

It improves the coefficient one in source `eq:block-square-root-bound`, using the same finite Laurent test degree. Although a particular block has $R \geq (1-2\lambda A)^n$, this lower bound can tend to zero when $n \asymp \delta/\lambda^2$ and $\lambda \to 0$. No uniformly useful positive floor is silently assumed.

4.2 Unit-disk centering is already available

The source's `eq:block-centering` states $|q| \leq 2$. Its following proof explicitly convolves β with an even nonnegative smooth approximate identity of integral one and observes that the resulting functions have absolute value at most one. We retain this stronger available conclusion:

$$|q| \leq 1, \quad q(-z) = \overline{q(z)}, \quad \int_{\mathbb{T}} \sqrt{p}\,|\beta - q|^2\,dz < \delta. \tag{4.7}$$

Indeed $|\beta| \leq 1$; convolution preserves the closed unit disk and the stated symmetry. At Lebesgue points it converges to β. Since $\sqrt{p} \in L^1$ and the squared difference is bounded by four, dominated convergence proves the weighted estimate. There is no claimed uniform bound on the derivatives of q, and q is fixed before the block frequencies.

Theorem 4.3 (Two explicit improvements of the ideal block loss)**.** *Suppose $p \geq 0$, $\int_{\mathbb{T}} p \leq 2$, and the block parameters and weighted centering condition above hold. With the source's weaker $|q| \leq 2$,*

$$\int \sqrt{p} - \int \sqrt{p}\,\mathbb{E}\sqrt{R} \leq [1 + 81\sqrt{2}e^{18\delta}]\delta^2 < 150\delta^2. \tag{4.8}$$

With (4.7)*, one has instead*

$$\int \sqrt{p} - \int \sqrt{p}\,\mathbb{E}\sqrt{R} \leq [1 + 16\sqrt{2}e^{8\delta}]\delta^2 < 26\delta^2. \tag{4.9}$$

For the first statement require $6\lambda < 1$; for the second $4\lambda < 1$ suffices, and retaining the source's $6\lambda < 1$ is harmless.

Proof. Use Lemma 4.1 with $A = 3$ or $A = 2$, multiply (4.6) by $\sqrt{p}$, and integrate. The first term is less than δ^2 by weighted centering. Cauchy–Schwarz on the unit-mass circle gives $\int \sqrt{p} \le \sqrt{2}$. For entirely rational upper estimates, $\sqrt{2} < 10/7$, and $e^x \le (1-x)^{-1}$ for $0 \le x < 1$, by comparing the nonnegative series termwise. Thus the constants are at most

$$1 + 81\frac{10}{7}\frac{50}{41} < 150, \qquad 1 + 16\frac{10}{7}\frac{25}{23} = \frac{4161}{161} < 26.$$

These prove the displayed strict bounds uniformly for $\delta \le 1/100$. □

Theorem 4.4 (Color-weighted refinement: ideal loss below three)**.** *Under the hypotheses of Theorem 4.3, with the source's original $|q| \le 2$ and $6\lambda < 1$, the ideal loss satisfies*

$$\int \sqrt{p} - \int \sqrt{p}\,\mathbb{E}\sqrt{R} \le [1 + e^{18\delta}(\sqrt{2} + 11\delta)]\delta^2 < 3\delta^2. \tag{4.10}$$

For the available unit-disk centering $|q| \le 1$, the first bracket can be replaced by $1 + e^{8\delta}(\sqrt{2} + 6\delta)$, also below three.

Proof. Put $e = \beta - q$ and $v = |e|^2$. The exact weighted variance identity is

$$S = \sum_r t_r |b_r - q|^2 = \sum_r t_r |b_r|^2 - |\beta|^2 + |\beta - q|^2 \le 1 + v. \tag{4.11}$$

For $|q| \le 2$, one has $v \le 9$, so $S^2 \le (1+v)^2 \le 1 + 11v$. For $|q| \le 1$, one instead has $v \le 4$ and $S^2 \le 1 + 6v$. Apply (4.4) and the sharp square-root minorant. The contribution of the first variance term is $\delta \int \sqrt{p}\, v < \delta^2$. The remaining contribution in the original case is at most

$$e^{18\delta}\delta^2 \int \sqrt{p}\,(1 + 11v) \le e^{18\delta}\delta^2(\sqrt{2} + 11\delta).$$

For the unit-disk choice replace $18, 11$ by $8, 6$. Finally the exact rational upper constants at $\delta \le 1/100$ are

$$1 + \frac{50}{41}\left(\frac{10}{7} + \frac{11}{100}\right) = \frac{1651}{574} < 3, \tag{4.12}$$

$$1 + \frac{25}{23}\left(\frac{10}{7} + \frac{6}{100}\right) = \frac{843}{322} < 3. \tag{4.13}$$

The bound $\sqrt{2} < 10/7$ and the exponential estimates were proved above. □

This improvement keeps the variation of the actual color amplitudes inside the Taylor remainder. It uses the same weighted centering already required by the source and introduces no new lower density bound, independence hypothesis, or geometric assumption. The number three is a convenient uniform bound, not a claim of the sharp possible block constant. The sharp optimality assertion concerns only the quadratic coefficient in Lemma 4.2.

4.3 Actual frequencies and actual error, not independent randomness

Source `lem:phase-tests` transfers a *finite list* of Laurent polynomials in $Z_1, \dots, Z_n$, with fixed L^1 coefficients in z, to the deterministic substitution $Z_i = e^{2\pi i L_i z}$ in integrated tests. Choose the frequencies successively so that every relevant nonzero integer combination escapes the finitely many

low frequencies of the coefficient functions. This is the Riemann–Lebesgue argument with finite Fourier approximation, not joint stochastic independence of the actual phases.

For the Hellinger estimate, apply that lemma to the polynomial $1 + (R-1)/2 - (R-1)^2/2$. Its coefficient functions are fixed by the pre-block densities, tags, centering function, n, and λ. They do not depend on future actual packets. The new coefficient $1/2$ and the smaller amplitude bound do not enlarge the finite test class or change its degree. The accumulated actual color-density error is tracked by

$$E_n \le M^n \eta_0 + \sum_{i=1}^{n} M^{n-i} \eta_i, \quad M = 1 + 2\lambda A. \tag{4.14}$$

Each finite error can be made smaller than $e_0/[(n+1)M^n]$ under the source realization lemmas. Such a requirement can be expensive, but is not circular: n, M, e_0 are fixed before the relevant large passage parameters.

If g and h are nonnegative circle densities, then

$$\left| \int \sqrt{g} - \int \sqrt{h} \right| \le \int \sqrt{|g-h|} \le \|g-h\|_1^{1/2}. \tag{4.15}$$

Consequently, if the integrated phase-test discrepancy is at most ξ and the final total-density discrepancy is at most e_0, the actual loss is bounded by $3\delta^2 + \xi + \sqrt{e_0}$. This explicit decomposition replaces an unspecified small remainder. Setting $\xi + \sqrt{e_0} \le \nu$ gives

$$\mathcal{A}_{k+1} \ge \mathcal{A}_k - 3\delta^2 - \nu_k, \qquad \mathcal{A}_k = \int \sqrt{p^{(k)}}. \tag{4.16}$$

4.4 Prediction variance and the accumulated improvement

Under the color-weighted phase law $H_r \, dm^n$, the estimator

$$D = q + \frac{1}{n\lambda} \sum_{i=1}^{n} Z_i$$

has expectation b_r and variance

$$\mathbb{E}_{H_r} |D - b_r|^2 = \frac{1 - \lambda^2 |a_r|^2}{n\lambda^2} \le \frac{2}{\delta}. \tag{4.17}$$

This follows from $\mathbb{E}_{H_r} Z_i = \lambda a_r$ and independence under the product law for each fixed color. The cross term with a fixed preceding estimator $A_k - k b_r$ vanishes at each z. Since $\int p \le 2$, each new block increases the integrated unnormalized prediction error by at most $4/\delta$ in the ideal model. The same finite phase-test argument and (4.14) control the actual version. With $|q| \le 1$ the deterministic estimator is bounded by $1 + 1/\lambda$, so a sufficient density error weight is

$$W_k = (\|A_k\|_\infty + 1 + 1/\lambda + k + 1)^2.$$

The source's larger weight remains valid. Nothing requires a uniform lower bound for λ over all blocks.

Corollary 4.5 (Conditional realized-block budget)**.** *Suppose the source's finite passage realization is available with the preceding physical transport, finite phase-test, and density-error budgets. Keep*

total mass at most two throughout, and use K blocks with $0 \le \nu_k \le \delta^2$ for both loss and prediction remainders. Then

$$\mathcal{A}_K \ge \mathcal{A}_0 - 3K\delta^2 - \sum_{k=0}^{K-1} \nu_k \ge \mathcal{A}_0 - 4K\delta^2, \tag{4.18}$$

$$\sum_r \int p_r^{(K)} |A_K/K - b_r|^2 \le \frac{4K/\delta + \sum_k \nu_k}{K^2} \le \frac{5}{K\delta}. \tag{4.19}$$

Proof. Sum (4.16). The prediction recurrence starts at $A_0 = 0$ with index zero, hence zero initial unnormalized error. Sum the increments and divide by K^2. Since $\delta \le 1$, $\delta^2 \le 1/\delta$, giving the last bound. □

The source's $300\delta^2$ ideal loss and $400K\delta^2$ total budget are therefore replaceable by the displayed constants under the same realization structure, with the already available centering choice. For example $K = \lceil 10/(\varepsilon^2\delta) \rceil$ gives prediction error squared at most $\varepsilon^2/2$, and it is enough to choose δ so that

$$4\left(\frac{10\delta}{\varepsilon^2} + \delta^2\right) < \tau. \tag{4.20}$$

Using only (4.9) would instead give $27K\delta^2$; the weighted refinement is what permits the stronger $4K\delta^2$ budget. The mass bound remains a separate physical condition: the source reserves total vector transport less than $\sqrt{2} - \sqrt{3/2}$ from an input of mass at most $3/2$. The numerical constants do not prove that reserve is met. Nor do they give an efficient explicit bound on the enormous possible frequencies, packet counts, or derivatives of the final diffeomorphism.

5 Whitening at density zero: existence, distance, and stability

Source `05-completion.tex`, `eq:whitening`, does not assume that the final pre-whitening density is positive almost everywhere. This is an important feature, not a gap to be filled by silently dividing by zero.

5.1 A flat vector already exists for each intermediate twist

Suppose $\phi' > 0$ on a nonempty arc I. Choose an integer $m > 0$ and $[u_0, u_1] \Subset I$ such that $m(\phi(u_1) - \phi(u_0)) = 1$. This is possible by choosing m large enough and then using continuity and strict monotonicity. Define

$$A(u) = \mathbf{1}_{[u_0,u_1]}(u)\sqrt{m\phi'(u)/2}, \qquad v = A(u)e^{2\pi i m\theta} + A(u)e^{2\pi i - m\theta}.$$

The two angular modes are orthogonal and their pushforward densities in (3.2) are each $1/2$ on the circle. Thus v is real, centered, has norm one, and $\sigma_v = m$. Its coefficients are L^2 with compact support in the good arc; they need not be smooth at the two endpoints. This is source `lem:flat-vector`. It supplies the actual spectral mass needed for the zero-set repair below, rather than an abstract promise that every operator has a flat vector.

Theorem 5.1 (Optimal flat-vector repair)**.** *Let U be a unitary on a complex Hilbert space and suppose $\sigma_G = p\,m$, with $p \ge 0$ integrable. Write $E = \{p = 0\}$, modulo m-null sets. Assume there is a vector v_E such that $\sigma_{v_E} = \mathbf{1}_E m$. Then*

$$F_* = (\mathbf{1}_{E^c} p^{-1/2})(U)G + v_E \tag{5.1}$$

is well-defined, $\sigma_{F_*} = m$*, and*

$$\|F_* - G\|^2 = \int_{\mathbb{T}} (\sqrt{p} - 1)^2 \, dm = \|G\|^2 + 1 - 2\int_{\mathbb{T}} \sqrt{p} \, dm. \tag{5.2}$$

It is a nearest vector to G *among all* F *with* $\sigma_F = m$*. The existence of some flat vector is equivalent to the availability of such a repair* v_E*, once* G *with the stated density is fixed.*

Proof. The multiplier in (5.1) is in $L^2(p\,m)$, since its squared norm is $m(E^c) \le 1$. Its image has spectral density $\mathbf{1}_{E^c}$. The spectral measure of v_E implies $E_U(E)v_E = v_E$, while $E_U(E)G = 0$. The two terms are spectrally orthogonal, so their densities add to one. The squared distance on E^c is $\int_{E^c}(1 - \sqrt{p})^2 \, dm$, and the squared norm of the repair on E is $m(E)$. This proves (5.2), including the zero set's contribution.

For optimality, let F have density one. The matrix of spectral measures of G, F is positive semidefinite, so the cross-measure has a density c with $|c|^2 \le p$ almost everywhere. One way to see this is to apply positivity to each rational complex linear combination of G, F and then use density of those coefficients to obtain pointwise positivity of the two-by-two density matrix. It follows that $\mathrm{Re}\langle G, F\rangle \le \int \sqrt{p}$. Expanding the squared distance gives the lower bound in (5.2). The constructed F_* attains it. Finally, a flat vector F gives a repair $E_U(E)F$; conversely a repair gives (5.1). This proves the equivalence. □

The hypothesis does not require v_E to be globally orthogonal to every packet of G. Only its spectral support on E is used. In M279 one takes $v_E = \mathbf{1}_E(U)v$ for the explicitly constructed flat vector v. If the Koopman operator commutes with complex conjugation, G and v are real, and $p(-z) = p(z)$, the multiplier and set obey the required reflection symmetry, and F_* is real. Under the twist formula (3.2), multipliers preserve the finite angular index set and good-arc support. The union with the two modes of v remains finite. This proves the admissibility facts needed at this particular completion step.

5.2 Why the zero repair cannot be omitted

Take a unitary model on $L^2(\mathbb{T}, m)$ and $G = \sqrt{2}\mathbf{1}_B$ with $m(B) = 1/2$. Then $\|G\| = 1$ and $p = 2\mathbf{1}_B$. Naive inverse-square-root multiplication produces $\mathbf{1}_B$, of squared norm $1/2$, not a unit flat vector. Taking $v = 1$ repairs it to $F_* = 1$ and gives squared distance $2 - \sqrt{2}$. If the Hilbert space were instead only $L^2(B, m)$, a repair on B^c would not exist at all. No transform inversion creates missing spectral support.

For any p, (5.2) implies

$$m\{p = 0\} \le \|F_* - G\|^2. \tag{5.3}$$

Near-unit mass alone gives no small whitening distance; the affinity $\int \sqrt{p}$ must also be close to one. This is the reason the block Hellinger estimate and physical transport budget both appear in the source.

Proposition 5.2 (A bounded regularization with its exact missing budget)**.** *Under Theorem 5.1, for* $\tau > 0$ *set*

$$F_\tau = (\max(p, \tau)^{-1/2})(U)G + v_E.$$

The multiplier on G *is bounded by* $\tau^{-1/2}$ *and*

$$\|F_\tau - F_*\|^2 = \int_{0<p<\tau} (1 - \sqrt{p/\tau})^2 \, dm \le m\{0 < p < \tau\}. \tag{5.4}$$

Hence $F_\tau \to F_*$ in norm as $\tau \downarrow 0$. If U, p, v_E are fixed and an approximate input $\widetilde{G}$ replaces G, then

$$\|\max(p,\tau)^{-1/2}(U)\widetilde{G} + v_E - F_*\| \le \tau^{-1/2}\|\widetilde{G} - G\| + m\{0 < p < \tau\}^{1/2}. \tag{5.5}$$

Proof. Subtract the two spectral multipliers. On $\{p \ge \tau\}$ they coincide; on $\{0 < p < \tau\}$ their squared difference multiplied by p is exactly the integrand in (5.4). Both repairs cancel. The positive low-density sets decrease to the empty set as $\tau \downarrow 0$; finite measure gives convergence. Bounded functional calculus and the triangle inequality prove (5.5). □

The last estimate is deliberately a *fixed-density, fixed-operator* statement. When applying the bounded multiplier to an arbitrary $\widetilde{G}$, fix a nonnegative finite Borel representative of p on the whole circle, including its m-null sets. Those values can matter for a singular spectral component of $\widetilde{G}$; the stated operator norm bound holds for any such fixed choice. The formulas on G and F_* are independent of that choice because their spectral measures are absolutely continuous. This proposition does not claim stability under arbitrary perturbations of p, its zero set, the repair vector, or U. A discontinuous spectral cutoff is not generally continuous under strong operator convergence. The regularized vector need not itself have exactly flat density. An executable procedure needs both a low-density mass certificate and a reconstruction error certificate; small pointwise sampling errors alone do not supply either.

5.3 Laurent prediction after whitening

In the source, a bounded multiplier D predicts the test function using G. Spectral calculus gives the exact identity

$$G = (\sqrt{p})(U)F_*, \qquad D(U)G = (D\sqrt{p})(U)F_*. \tag{5.6}$$

The multiplier $D\sqrt{p}$ belongs to $L^2(m)$, since D is bounded and $p \in L^1$. Laurent polynomials are dense in that space. Because $\sigma_{F_*} = m$, approximating $D\sqrt{p}$ by a Laurent polynomial P gives exactly the same L^2 error for $P(U)F_*$. There is no division of the prediction error by an unknown lower density bound. Nor is a bound on the degree or coefficient size of P asserted. The polynomial is chosen after whitening and is then fixed for all future retention conditions.

6 Physical Gram certificates for moving orbit windows

This section gives a direct refinement of TR-item-169, Section 14. All operators and norms refer to the same physical Hilbert space. Changing coordinates without transporting its Gram metric would invalidate the statements.

6.1 Exact synthesis and residual formulas

Let $v_1, \dots, v_d \in \mathcal{H}$ be independent columns and let $Jc = \sum_{j=1}^{d} c_j v_j$. With our linear-first convention,

$$G = J^*J, \quad G_{jk} = \langle v_k, v_j\rangle, \quad \|Jc\|^2 = c^*Gc, \quad \Pi = JG^{-1}J^*. \tag{6.1}$$

The transpose placement in the displayed inner product is essential. Π is the physical orthogonal projector onto $\operatorname{ran} J$. For a target h, put $b = J^*h$ and $c_* = G^{-1}b$. Then

$$\|h - Jc\|^2 = \|h\|^2 - b^*G^{-1}b + (c - c_*)^*G(c - c_*). \tag{6.2}$$

To prove it, decompose $h - Jc = (I - \Pi)h + J(c_* - c)$ into orthogonal terms. This is the finite prediction version of the Volume 21 physical residual identity, with the unrepresented component explicitly retained. A zero normal-equation residual is not evidence that h lies in $\operatorname{ran} J$.

For orbit columns $v_k = U^k f$ on Λ, the entries are $G_{jk} = r_{k-j}$, where $r_\ell = \langle U^\ell f, f\rangle$. If $\sigma_f = m$, then

$$G = I \quad \text{on every finite orbit window.} \tag{6.3}$$

This is a frame bound on that window's span, not a completeness theorem on the whole Hilbert space. In M279 it holds exactly at every admissible stage.

Proposition 6.1 (Correlation and matrix-enclosure certificates)**.** *Suppose $\|f\| = 1$ and Λ is a finite set of distinct integers. Define*

$$\rho_\Lambda = \max_{j\in\Lambda} \sum_{\substack{k\in\Lambda \\ k\neq j}} |r_{k-j}|.$$

Then $(1 - \rho_\Lambda)I \preceq G \preceq (1 + \rho_\Lambda)I$. For $\Lambda = \{-K, \dots, K\}$ a sufficient simpler bound is

$$G \succeq \left(1 - 2K \max_{1\le|\ell|\le 2K} |r_\ell|\right) I. \tag{6.4}$$

If a Hermitian approximation $\widehat{G}$ satisfies $\|G - \widehat{G}\| \le \eta$ and an exact factorization certifies $\widehat{G} - (a + \eta)I \succeq 0$, then $G \succeq aI$.

Proof. The Hermitian matrix $G - I$ has operator norm bounded by its maximal absolute row sum, yielding the first conclusion. Each row has $2K$ off-diagonal entries, whose differences lie in $[-2K, 2K]$, proving (6.4). For the last assertion, $G \succeq \widehat{G} - \eta I \succeq aI$. □

Thus the correlation window is $2K$, not K. A rational LDL^* factorization with strictly positive diagonal pivots is one exact positive-definiteness certificate. A floating-point eigenvalue alone is not such a certificate. Errors in quadrature, physical mass, and stored correlations must enter η. Revised TR248, `thm:perturb`, provides the corresponding singular-value margin $(\sqrt{A} - \varepsilon)^2$ for a perturbed synthesis or analysis map whose operator error is bounded by $\varepsilon < \sqrt{A}$.

Theorem 6.2 (Gram-normalized physical window transfer)**.** *Let $J, L : \mathbb{C}^d \to \mathcal{H}$ be bounded linear maps, with J injective, $G = J^*J$, and*

$$\eta = \|(J - L)G^{-1/2}\|. \tag{6.5}$$

For every $h \in \mathcal{H}$,

$$\operatorname{dist}(h, \operatorname{ran} L) \le \|(I - \Pi)h\| + \eta\|\Pi h\|, \qquad \Pi = JG^{-1}J^*. \tag{6.6}$$

The number η is invariant under any common invertible reparameterization $J \mapsto JA$, $L \mapsto LA$. Equivalently,

$$\eta^2 = \sup_{c\neq 0} \frac{c^*(J - L)^*(J - L)c}{c^*Gc}. \tag{6.7}$$

Proof. Use $c = G^{-1}J^*h$. Then $Jc = \Pi h$ and $\|G^{1/2}c\| = \|\Pi h\|$. The triangle inequality applied to $h - Lc = (h - Jc) + (J - L)c$ proves (6.6). The variational quotient is precisely the squared operator norm in (6.5). Under the common reparameterization its numerator and denominator undergo the same congruence. Substitution $c = Ad$ leaves the supremum unchanged. □

The bound from TR-item-169 follows because $\eta \le \|J - L\|/\sqrt{a}$ whenever $G \succeq aI$. The normalized version can be strictly sharper and does not penalize a harmless poorly scaled coefficient basis. It still requires a bound on the *physical* transfer error, not only the Gram matrix of the approximant. For the actual Haar windows in M279, $G = I$, so there is no small lower-frame denominator to cure. The challenge is the moving-window error itself.

6.2 A directional certificate for a chosen predictor

Let $e_k = v_k - w_k$ and let the error Gram matrix be $R_{jk} = \langle e_k, e_j \rangle$. For a particular coefficient vector c,

$$\|\sum_k c_k e_k\|^2 = c^* R c. \tag{6.8}$$

This exact positive quadratic form can be much smaller than $\|R\|\|c\|^2$. If $\|e_k\| \le \zeta_k$, then

$$\|\sum_k c_k e_k\| \le \sum_k |c_k|\zeta_k \le \|c\|_2 \left(\sum_k \zeta_k^2\right)^{1/2}. \tag{6.9}$$

One may certify the error for selected predictions instead of all vectors in an ever-growing window. Source `eq:retain-polynomial` makes exactly this targeted choice. Its compatibility with finite continuity conditions is more important than inventing an unproved uniform frame-transfer rate.

6.3 Why a finite Ram panel cannot certify the whole orbit space

The finite-window projector has an orthogonal complement unless its range is all of $\mathcal{H}$. A complete boundary-local Ram bank has a Parseval identity, but finite truncation introduces the exact positive leakage operator

$$L_{\mathrm{leak}} = \sum_p M_{w_p}(I - \Pi_p)M_{w_p}, \qquad F^*F = I - L_{\mathrm{leak}}.$$

On a named finite model V with isometric synthesis Q, its lower bound is $1 - \|Q^* L_{\mathrm{leak}} Q\|$. This is useful for implementing a finite projection of a packet or target. It does not prove that the orbit span is dense as both the map and the model change. The exact missing quantity is the physical best-approximation residual in (6.2), plus the transfer error in (6.6). This is the concrete scope of the foundation frame results in M279.

7 Passing to the smooth limit without losing cyclicity

7.1 An explicit counterexample to fixed-power reasoning alone

Proposition 7.1 (Haar orthonormal bases can lose completeness in a strong limit)**.** *There are unitaries U_j, U on a fixed separable Hilbert space and a fixed unit vector f such that $U_j^n f \to U^n f$ for every fixed integer n, indeed $U_j^n \to U^n$ strongly, and every orbit $\{U_j^n f : n \in \mathbb{Z}\}$ is an orthonormal basis. The limiting orbit is orthonormal but not complete. This remains possible with a moving predictor having just one coefficient of modulus one.*

Proof. Take $\mathcal{H} = \ell^2(\mathbb{Z} \times \{1,2\})$ with basis $e_{n,r}$, let $Ue_{n,r} = e_{n+1,r}$, and set $f = e_{0,1}$. For each $j \geq 1$ choose a bijection $a_j : \mathbb{Z} \to \mathbb{Z} \times \{1,2\}$ satisfying

$$\begin{aligned} a_j(k) &= (k,1) && (-j \leq k \leq j), \\ a_j(k) &= (k-(4j+1),2) && (3j+1 \leq k \leq 5j+1). \end{aligned}$$

These prescriptions use disjoint finite domain intervals and disjoint finite images. Extend arbitrarily by a bijection of their countably infinite complements. Define the single bilateral permutation shift $U_j e_{a_j(k)} = e_{a_j(k+1)}$. Its orbit from $f = e_{a_j(0)}$ is the entire orthonormal basis. For any fixed (n,r), both its predecessor and successor eventually lie in the prescribed interior of the appropriate block. Therefore U_j and U_j^{-1} agree with U and U^{-1} on each fixed basis vector for all sufficiently large j. Density of finite-support vectors and unitarity imply strong convergence of every fixed positive or negative power.

Nevertheless, the limiting cyclic space of f is only the first copy. For $h = e_{0,2}$, the moving polynomial $P_j(z) = z^{4j+1}$ satisfies $P_j(U_j)f = h$, while $P_j(U)f = e_{4j+1,1}$ is orthogonal to h. The error has norm $\sqrt{2}$, even though its coefficient vector has norm one and every finite approximating orbit Gram matrix is the identity. □

This is an abstract Hilbert-space counterexample to an inference, not a counterexample to M279's geometric construction or to its extra retention conditions. It shows why exact Haar correlations and bounded coefficient norms still do not control moving Laurent degrees.

Theorem 7.2 (Physical moving-window completion)**.** *Let U_j, U be unitaries on one Hilbert space, with $U_j^n \to U^n$ strongly for every fixed $n \in \mathbb{Z}$, and let unit vectors $f_j \to f$. Suppose their correlations converge to Haar, and choose finite windows Λ_j with injective synthesis*

$$J_j c = \sum_{n \in \Lambda_j} c_n U_j^n f_j, \qquad L_j c = \sum_{n \in \Lambda_j} c_n U^n f.$$

Assume $\eta_j = \|(J_j - L_j)(J_j^* J_j)^{-1/2}\| \to 0$. If every member of a family with dense linear span is approximated by $\operatorname{ran} J_j$ *with distance tending to zero, then $\{U^n f : n \in \mathbb{Z}\}$ is an orthonormal basis.*

Proof. Fixed-power strong convergence and $f_j \to f$ pass each scalar correlation to the limit, giving $\langle U^n f, f\rangle = \mathbf{1}_{n=0}$. Hence the orbit is orthonormal. For a fixed target h from the dense family, Theorem 6.2 gives $\operatorname{dist}(h, \operatorname{ran} L_j) \leq \operatorname{dist}(h, \operatorname{ran} J_j) + \eta_j \|h\| \to 0$. Every $\operatorname{ran} L_j$ lies in the closed cyclic subspace of f, so that subspace contains the dense family and equals $\mathcal{H}$. □

This restates TR-item-169's corrected conclusion with the sharper Gram-relative transfer norm. In the special window $[-K_j, K_j]$, the local report's sufficient budget is $\sqrt{2K_j+1}\,\zeta_j/\sqrt{a_j} \to 0$ for a uniform column error ζ_j and lower Gram bound a_j. Mere convergence of each individual column does not imply that growing-window budget.

7.2 The targeted criterion actually used by M279

Theorem 7.3 (Retention of chosen Laurent predictions)**.** *Suppose $U_j^n \to U^n$ strongly for fixed integers n, $f_j \to f$, and $\sigma_{f_j} = m$. Let (h_j) be a sequence whose members have dense linear span and in which each member of a fixed generating family occurs infinitely often. Suppose finite Laurent polynomials P_j satisfy*

$$\|h_j - P_j(U_j) f_j\| \leq \alpha_j, \quad \|P_j(U_n) f_n - P_j(U_{n-1}) f_{n-1}\| \leq b_{j,n} \quad (n > j), \tag{7.1}$$

where $\sum_{n>j} b_{j,n}$ is finite. If $\alpha_j + \sum_{n>j} b_{j,n} \to 0$ along the occurrences of each fixed generating target, then f has a complete Haar orbit for U.

Proof. For each fixed j, telescoping gives

$$\|h_j - P_j(U_N) f_N\| \le \alpha_j + \sum_{n=j+1}^{N} b_{j,n}.$$

Because P_j has finitely many fixed powers, strong convergence and vector convergence imply $P_j(U_N) f_N \to P_j(U) f$. Pass to the limit, and then take repeated occurrences of each target. They all belong to the closed cyclic subspace of f. Haar correlations pass to the limit by the same fixed-power argument, giving orthonormality. □

No bound uniform in j is required on the degrees or coefficients of P_j. At the nth stage there are only $n-1$ old polynomials to retain. If $P(z) = \sum_{k\in\Lambda} c_k z^k$ is fixed, then

$$\|P(U')g' - P(U)g\| \le \sum_{k\in\Lambda} |c_k| (\|g' - g\| + \|(U'^k - U^k)g\|). \tag{7.2}$$

Finite coefficients, however large, admit a sufficiently small neighborhood. There is no passage from fixed-power continuity to an unproved uniform claim over all later polynomials.

Selected retention is strictly weaker than whole-window transfer. The independent source audit supplies an additional exact example on $\ell^2(\mathbb{Z})$. Let $Ue_k = e_{k+1}$ and let $U_j e_k = e_{k+1}$ except that $U_j e_j = -e_{j+1}$. These are bilateral weighted shifts with unit-modulus weights. With $f = e_0$, both the limiting and approximating orbits are complete and exactly Haar. Every fixed Laurent prediction agrees eventually, and fixed powers converge strongly. But on $[-j-1, j+1]$ the column at $n = j+1$ differs by norm two. All other columns in that window agree, so the synthesis-difference operator norm is exactly two, not a quantity tending to zero. Thus even successful selected predictions and a complete limiting orbit do not imply the stronger moving-window hypothesis.

One can explicitly impose that stronger hypothesis if it is useful. After selecting a finite stage window Λ_j, retain each of its columns through every later stage with a summable total error at most ζ_j. At any new stage there are finitely many old columns to retain. Their continuity conditions are compatible with the source's arbitrarily small physical stage perturbations. Then

$$\|J_j - L_j\| \le \sqrt{|\Lambda_j|}\, \zeta_j,$$

so choosing $\sqrt{|\Lambda_j|}\, \zeta_j \to 0$ enforces the window criterion when $G_j = I$. This is an *additional* ledger of finite constraints, conditional on the same one-stage realization. It is not a hypothesis that the source's selected-polynomial proof already established automatically.

7.3 Smooth physical convergence and the source's explicit tail

For volume-preserving maps $S_n \to T$ and inverse maps $S_n^{-1} \to T^{-1}$ uniformly on a compact manifold, every fixed iterate converges uniformly. For continuous g, uniform continuity gives $g \circ S_n^k \to g \circ T^k$ uniformly. Approximate an arbitrary L^2 function by continuous functions; unitarity controls the two approximation errors. This proves fixed-power strong Koopman convergence on the actual physical space, including negative powers. It does not require a smooth representative for the flat vector.

The source chooses successive maps and inverses Cauchy in all C^k norms by imposing summable errors for $k \le n$. Their limits are smooth inverse maps, because the inverse identities pass to the uniform limit. Volume preservation passes through integration of continuous tests. The L^2 increments of the flat vectors are summable, so their norm-one, real, centered limit exists in the same physical space.

Source `prop:stage` gives a prediction error $\sqrt{\beta_{j-1}} + 2^{-j}$, where β_{j-1} is the previous bad-arc length, and source `eq:retain-polynomial` takes $b_{j,n} < 2^{-n}$. For $j \ge 2$, $\beta_{j-1} < 4^{-(j-1)}$, so

$$\|h_j - P_j(U_T)f\| \le \sqrt{\beta_{j-1}} + 2^{-j} + \sum_{n>j} 2^{-n} \le 2^{2-j}. \tag{7.3}$$

Thus the completion argument does contain the missing physical retention condition emphasized by TR-item-169. This is a positive bounded finding. It is not a proof that all preceding passages realize the one-stage lemma with all the imposed smallness conditions.

8 What the one-stage construction still has to supply

The source stage begins with an admissible flat vector of finite angular support in an intermediate twist chart. Given a real smooth centered target h with $\|h\|_\infty \le 1$, it seeks a nearby admissible pair and one Laurent prediction. This subsection records why the target is not merely a packet, and where the nontrivial geometric work remains.

First decompose h into its angular mean and mean-zero angular part. Cutting the latter off outside a good arc of complement length β costs at most $\sqrt{\beta}$ in L^2, by fiberwise Parseval and $\|h\|_\infty \le 1$. The two remaining successive means depend first on (u, a) and then on a alone. Two neutral passages rotate those old coordinates into angular coordinates. The finite Fourier cutoff of each target component is fixed before the next large passage parameter; old packet indices are then separated from those finite low indices. This addresses the full centered target space, not only observables that already have zero angular mean in one chart.

Next the source approximates the current flat vector by a smooth packet sum F and the target by another packet sum H. On each shared angular mode, an arbitrarily small real perturbation $F + tH$ can be chosen so that H_m/F_m is defined almost everywhere with respect to $|F_m|^2\,du\,da$. For almost every t, the set where $F_m + tH_m = 0$ but $H_m \neq 0$ has zero measure: each physical point permits at most one such real t, and Fubini applies. The weighted identity

$$\int \left|\frac{H_m}{F_m}\right|^2 |F_m|^2 = \int |H_m|^2$$

then justifies truncating the ratio in the weighted measure and approximating it by finite constant tags on rectangles, with smooth interior cutoffs. The resulting tag bound can be very large. It must be fixed before choosing later transport and prediction tolerances.

Two more neutral passages separate these colors by angular index alone and make their coefficient moduli independent of the new spare coordinate. Those are exactly the hypotheses of the signal and prediction construction. The physical transport errors are divided by a factor depending on the maximum tag so that both the untagged and tagged fields remain close. At the end, near-unit mass and large Hellinger affinity permit Theorem 5.1; (5.6) then replaces the bounded multiplier by a finite Laurent polynomial.

Each ingredient has a stated source location, and the particular zero-set and fixed-polynomial objections are addressed explicitly. Remaining independent verification would have to cover, among

other things, the uniform summed stationary-phase comparison, all cutoff losses, law approximation after periodization, simultaneous finite-list transport, support separation after both coordinate rotations, and the complete parameter order for physical C^k neighborhoods. None of these follows from the sharp scalar constant or a positive finite Gram matrix.

9 Consequences with their exact hypotheses

Proposition 9.1 (A finite-tail ordinary-mixing certificate)**.** *Suppose a unitary U on $\mathcal{H}$ has a complete orthonormal orbit $(U^k f)_{k\in\mathbb{Z}}$. Let a_K and b_L be the orthogonal projections of $a, b \in \mathcal{H}$ onto the windows $[-K, K]$ and $[-L, L]$. For $|n| > K + L$,*

$$|\langle U^n a, b\rangle| \le \|a - a_K\|\|b\| + \|a_K\|\|b - b_L\|. \tag{9.1}$$

In particular all centered correlations tend to zero.

Proof. The shifted finite orbit support of a_K does not meet that of b_L, so $\langle U^n a_K, b_L\rangle = 0$. Subtract this zero correlation, split the remaining two terms, and use unitarity and Cauchy–Schwarz. Completeness makes the projection tails tend to zero; first choose fixed K, L and then let $|n| > K+L$. □

For the Koopman realization, applying this to centered indicator functions proves ordinary measure-theoretic mixing and hence ergodicity. The bound is a useful conditional extension: it makes the needed approximation tails explicit. It does not imply a numerical decay rate for smooth observables without a quantitative relation between their physical smoothness and their coefficients in this potentially rough orbit basis.

The source's zero-entropy conclusion invokes the classical implication that a probability-preserving transformation of finite spectral multiplicity has zero Kolmogorov–Sinai entropy; source `06-consequences.tex` cites Rokhlin's 1960 account [12]. Given that theorem and the claimed global construction, Pesin's entropy formula for smooth volume-preserving diffeomorphisms, in the form cited by the source from Mañé [13], gives zero sum of positive Lyapunov exponents almost everywhere. Volume preservation gives zero sum of all exponents, so all exponents vanish. This uses smooth compact dynamics and the absolutely continuous invariant measure, not spectral algebra alone. The relevant entropy and Pesin proofs were not independently reproduced here.

The all-orders-mixing assertion in the source depends on another public OpenAI manuscript, *Rokhlin's multiple-mixing problem for one transformation*, Theorem 1.1 [14]. It is not audited here. No theorem above upgrades pairwise mixing to all orders: the unitary spectral model need not preserve products of observables, whereas multiple mixing involves such products. The conclusion must remain conditional on that additional theorem, rather than being presented as an independently established refinement of M279.

10 Evidence, limits, and practical next steps

10.1 Reading and verification depth

The entire pinned mathematical source of M279 was read: all six section files, its main file, and bibliography. That includes the passage, signal, block, stage, completion, and consequence arguments, not only titles or introduction. TR-item-169 was read in full. The specific frame and metric sections

listed in Appendix A were read in the actual local files. Corpus maps were used to locate those files, not substituted for their hypotheses or proofs.

This report proves its scalar and operator lemmas directly. Finite exact diagnostics are supplementary checks of formulas, not certificates of all parameters or of an infinite smooth limit. The local record `C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/check_batch3_parent.py` and its same-stem JSON were read. The M279 portion reports 404 successful exact rational evaluations of the factorization in Lemma 4.2. The other manuscripts' diagnostics in that shared file are not evidence about M279 and are not used here. The M279 factorization has the symbolic proof given above; testing 404 inputs is neither necessary nor sufficient for its universal quantifier.

The independent source audit at `C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/M279-source-audit.txt` was read, including its sections A–J and the later refinement in Section L. It independently reconstructs the complex block moments, zero-set repair, and fixed-polynomial telescoping, and records no established fatal counterexample or circularity in the source chain at its stated depth. Its principal mapping caution is incorporated here: retention of selected predictions does not imply operator-norm transfer of every growing orbit window. The two sufficient criteria are therefore stated separately in Section 7. Its scalar refinement uses the weaker centering bound and obtains $150\delta^2$; the further $26\delta^2$ intermediate result here also uses the explicit source choice $|q| \le 1$. Section L of the independent audit separately checks the still stronger weighted-amplitude refinement $3\delta^2$, including the exact $1651/574$ and $843/322$ constants and the $4K\delta^2$ accumulation. The complete proof is retained in Theorem 4.4. Section M of the audit subsequently checked the displayed local variance, whitening, regularization, physical Gram, counterexample, retention, and mixing propositions under their stated hypotheses. Its minor notation correction was applied, and its Borel-representative convention for a possibly singular approximate input is explicit in Section 5. That follow-up reviewed an earlier complete text and the later mathematical changes; it is not described as a frozen-byte or PDF audit of this final file. Neither full-source reading nor the audit's negative finding is treated as global theorem certification. The audit's external entropy references were not fully verified there, and they are not promoted to full reads here.

The local report checker `C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/check_M279_report_local.py` was executed successfully with the bundled Python runtime; its saved same-stem JSON records 353 finite cases using exact rational arithmetic. These comprise 164 square-root evaluations, five rational constant checks, six complex finite-phase experiments, nine weighted-color variance cases, 25 physical-matrix cases, 64 piecewise-density whitening cases, and 80 diagonal/counterexample cases. Each phase experiment checks all its color means, cross moments, estimator variances, and a rational variance upper bound. Four equally spaced phases integrate the finite Laurent degrees in those experiments exactly; this is not a Monte Carlo check. The counterexample tests check its finite prescriptions, not the abstract infinite extension by enumeration.

The independent audit's checker `C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/check_M279_source_local.py` was also read and independently rerun successfully. It checks the sharp factorization by polynomial coefficient identity, the rational $6116/41 < 150$ bound, 72 real-amplitude moment cases, 3,006 prime-sieve instances, a zero-set example, and 100 weighted-shift windows. None of these checks implements a final torus map, verifies all analytic cutoffs, or proves the imported entropy or multiple-mixing results. The report's proofs, not the case counts, support its universal local statements.

The code and saved result of the separate check `C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/check_M279_centered_mass.py` and its same-stem JSON were also read. That record reports 24,843 exact rational two-color fixtures checking (4.11), both squared amplitude

bounds, and the constants in (4.13). This execution was independently supplied; it is not counted as an additional rerun here. The analytic pairwise Taylor and weighted integral argument in Theorem 4.4, rather than those fixtures, proves the refinement for arbitrarily many colors.

10.2 Advantages and costs of the concrete extensions

The $3\delta^2$ ideal and $4K\delta^2$ actual block estimates reduce the permitted Hellinger loss without introducing a new geometric assumption. They keep actual phase and density errors explicit and preserve the source's prediction-variance argument. Their limitation is equally concrete: a smaller loss coefficient does not provide a practical upper bound on passage size or derivative growth.

The optimal whitening theorem gives an exact distance, a necessary spectral availability condition for the zero repair, and a bounded regularization with a low-density remainder. This is stronger and more transparent than formal division by $\sqrt{p}$. Its implementation still needs control of small spectral densities and actual spectral projections. These can be unstable or unavailable numerically.

The Gram-relative transfer theorem is coordinate-invariant and can exploit anisotropic error better than a global lower-frame denominator. The directional error Gram may reduce the cost further for selected targets. But dense matrix certification costs memory and arithmetic, and finite projection leaves an off-model residual. When the stage vectors already have exact Haar density, the Gram is already identity; the framework cannot improve a nonexistent conditioning problem. Its value then is to expose the physical transfer error and explain the source's retention mechanism.

The abstract counterexample gives a precise warning even in an ideally conditioned setting. It neither diagnoses a flaw in the source nor replaces an actual chart-level analysis. In particular, the source has stronger retention hypotheses than those violated by the example.

10.3 A focused future research program

1. Produce a self-contained uniform stationary-phase appendix for the exact compact packet class, with summed coefficient and density errors, all derivative constants, and nonstationary tail summation. This is a higher-priority validation step than testing more scalar inequalities.
2. Turn the ordered choices of cutoffs, c, jitter width, N, tags, and physical C^k neighborhoods into a finite-stage certificate ledger. Record which constants are fixed before each later choice. The number of passages per block must be known before allocating its movement budget.
3. Investigate low-density mass estimates strong enough to make (5.5) effective. A lower bound on a sampled density is not a substitute for a continuum bound or a certified exceptional set.
4. Certify retained predictions in the actual physical chart using the directional error Gram (6.8). Compare this with the coarser coefficient ℓ^1 budget at the same rigorous error level.
5. Quantify the projection tails in (9.1) for an explicit class of physical observables. Do not infer smooth-observable rates from smoothness of T when the flat cyclic vector is only L^2.
6. Audit the independent entropy and multiple-mixing dependencies on their own stated hypotheses. An all-orders conclusion should not be added to a local block refinement merely because it appears as a source corollary.

10.4 Bottom line

The Ram corpus is directly relevant to the core problem because its corrected Koopman criterion names exactly the Haar-correlation, full-cyclicity, and physical-limit obligations. The pinned source explicitly handles density zeros and the moving-prediction issue; no defect in either step is inferred from their abstract pitfalls. The strongest concrete improvement established here is the smaller Hellinger budget, with the zero-repair and physical Gram theorems providing rigorous complementary extensions. Acceptance of the global smooth Banach construction remains beyond the independent analytic verification performed for this report.

A Source manifest and exact locators

A.1 Pinned public manuscript and actual reading

The exact catalogue row used was M279 in `C:/Users/mural/OneDrive/Documents/OpenAI-722-math/00-Applicability-Catalogue/applicability.json`. It identifies the public manuscript by OpenAI, dated September 23, 2026, and the commit

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

The pinned public PDF URL is given in reference [1]. The mathematical source actually read is the local download at the following absolute base directory, denoted S only to keep the locator table legible:

> `C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_sources/openai-math/preprints/A-smooth-three-torus-diffeomorphism-with-simple-Lebesgue-spectrum-September-23-2026/build`

Every filename in the table is appended literally to S. These are public source copies, not confidential Ram corpus documents. The local files were not authenticated against a Git checkout by this report; the catalogue supplies the pin and the independent audit records local SHA-256 fingerprints.

File under S	Exact locations and actual reading depth
`main.tex`; `references.bib`	Fully read. Main definition of the claimed result, authorship/date, six includes, and the external dependency bibliography.
`sections/01-introduction.tex`	Fully read, lines 1–144. `thm:main`; full centered-space claim, prior-result distinctions, and construction order.
`sections/02-passages.tex`	Fully read, lines 1–505. `eq:packet-spectral-measure`, `eq:density-norm-bound`, `lem:passage`, `eq:summed-stationary-mass`, `eq:rounding-mass-removal`, `eq:K-jacobian`, `lem:fast-average`, `lem:law-approx`, `lem:neutral`, `rem:finite-family-passages`.
`sections/03-signals.tex`	Fully read, lines 1–459. `lem:signal`; positive law `eq:signal-positive-law`; Fourier identity `eq:signal-joint-fourier`; `eq:signal-finite-c`; both drift equations; `lem:refresh-transmit` and `eq:signal-refresh-error`.

File under S	Exact locations and actual reading depth
`sections/04-blocks.tex`	Fully read, lines 1–511. `prop:prediction`; `lem:phase-tests`; centering and following convolution proof, lines 123–142; `eq:block-second-moment-identity`; `eq:block-second-moment-bound`; `eq:block-square-root-bound`; `eq:block-estimator-moments`; `eq:block-total-hellinger`; `eq:block-total-prediction`.
`sections/05-completion.tex`	Fully read, lines 1–440. `lem:flat-vector`; `prop:stage`; angular averages and tag preparation, lines 76–262; `eq:whitening`; `eq:whitening-distance`; `eq:strong-koopman`; `eq:retain-polynomial`; `eq:limit-orthonormal`.
`sections/06-consequences.tex`	Fully read, lines 1–65. `cor:dynamical-profile`; ordinary mixing, imported higher-order theorem, and conditional entropy/Pesin chain.

The six substantive section files contain 2,124 lines in the inspected snapshot. Full reading is a statement about coverage, not proof completeness. The uniform analytic families were traced rather than reconstructed as a separate all-parameter implementation. No original source was edited.

A.2 Confidential local corpus: actual files, not just maps

R169. `C:/Users/mural/OneDrive/Documents/My Documents 2026/res-rt/res-math/TR-item-169-astra.tex`

Read in full, including the corrected later sections rather than only the earlier provisional criterion. The exact section titles used are *Authoritative formulation* (Section 1), *Equivalent correlation criterion* (Section 3), *Finite-tower completion theorem* (Section 6), *Audit result and corrected source boundaries* (Section 11), *Two elementary impossibility tests* (Section 12), *A valid replacement for global uniform flatness* (Section 13), *Moving-window cyclicity with a physical frame budget* (Section 14), and *Mass, flow restrictions, and numerical certification* (Section 15). Section 14's theorem is titled *Corrected finite-to-infinite cyclicity theorem*; its hypothesis labels are `eq:cor169`, `eq:dense169`, and `eq:transfer169`. The source's September 7 historical status statement predates M279; it is not used to refute or certify a later claim. Its exceptional-set logarithm theorem requires uniform integrability of the *products*; factorwise control alone is not a substitute. That alternative Riesz-product route is not needed for the present block proof.

R248. `C:/Users/mural/OneDrive/Documents/RamFoundationTRs/TR-rev-1-old-tr248/TR-rev-1-old-tr248.tex`

Actual sections read and used: *Spaces, measurements, and the exactness boundary*; *E1: boundary-local frames with exact leakage accounting*; *E2: implementation-stable frame certificates*; and appendix *Moment assembly, analytic tails, and exact certificates*. Exact labels: `eq:ramatom`, `thm:parseval`, `eq:leak`, `thm:finite`, `eq:modelerror`, `thm:perturb`, `eq:momentgram`, and `eq:momenttail`. Other sections were encountered during the read but are not claimed as independently audited. The finite experiments archived with R248 were not rerun for M279.

V2. `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume2/chapters/ch02-frames-duals.tex`

Actual sections used: `v2:c02:geometry`, `v2:c02:frame-operators`, `v2:c02:duals`, `v2:c02:finite`, `v2:c02:patches`, `v2:c02:leakage`, and `v2:c02:perturbation`. The stable inversion theorem, `v2:c02:canonical-decoder`, `v2:c02:coefficient-projector`, `v2:c02:mass`, and `v2:c02:model-error` were read with proofs. This is targeted chapter-section reading, not a claim to have audited every later exercise, noise model, or application in the long chapter.

V21. `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/ch01d-physical-metrics.tex`

The whole chapter fragment was read. Exact labels used: `v21c01:physical-gram-definition`, `v21c01:metric-symmetry`, `v21c01:residual-decomposition`, `v21c01:residual-identity`, `v21c01:gram-equivalence`, and `v21c01:finite-measurements`. Its column formulas are translated to the linear-first convention explicitly used in this report.

Navigation maps. Targeted relevant entries were searched in the following local files:

```
C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/corpus-foundations-map.txt
C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/corpus-core-map.txt
C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/corpus-applications-map.txt
```

No map entry is used as a replacement for an original theorem. No confidential corpus text or proposed private construction was submitted to web search.

A.3 Public-reference access depth

Ryzhikov's Section 1 and Q2 were read in the public HTML version. Prikhod'ko's MathNet abstract and publication metadata were checked; its full proof was not reread. For Fayad–Forni–Kanigowski the author-hosted PDF's Theorem 1, contextual discussion, and Section 7 criterion were examined, not all 66 pages. The Abdedou–Fayad–Kessi publisher abstract and publication record and the Abdedou–Fayad–Thouvenot primary abstract/publication record were checked. The el Abdalaoui preprint record was checked without accepting its global proof as an input. Rokhlin's MathNet bibliographic record was accessible, but not the cited theorem text through that fetch. Direct Mañé DOI fetches failed; the title and citation occur in the local source bibliography and a primary publisher reference list. The entropy and Pesin theorems remain named external imports. The separate OpenAI multiple-mixing manuscript was not read as part of this batch task.

References

[1] OpenAI. *A smooth three-torus diffeomorphism with simple Lebesgue spectrum.* September 23, 2026. Public preprint, catalogue M279. Pinned source: `https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/A-smooth-three-torus-diffeomorphism-with-simple-Lebesgue-spectrum-September-23-2026/paper.pdf`. Local source provenance and exact labels are in Appendix A.

[2] Local Ram corpus. *Ram-Calculus Spectral Certificates for Banach's Simple Lebesgue Spectrum Problem.* TR-item-169-astra, September 7, 2026. Exact local file R169 in Appendix A. Confidential local source; no public dissemination inferred.

[3] M. Subbarao. *Stable Adaptive Ram Frames and Task-Aware Approximation*. Foundational revision TR-rev-1-old-tr248 of TR248, September 2026. Exact local file R248 in Appendix A. Confidential local source.

[4] Ram Calculus corpus. Volume 2, Chapter 2, *Frames, Duals, and Finite Measurement Models*. Exact local file V2 and used section labels in Appendix A. Confidential local source.

[5] Ram Calculus corpus. Volume 21, Chapter 1 physical-metric fragment, *A Nonorthogonal Representation Carries a Metric* and following sections. Exact local file V21 in Appendix A. Confidential local source.

[6] V. V. Ryzhikov. *Unsolved problems on joinings, multiple mixing, spectrum, and rank*. 2024, arXiv:2405.03641v1. Section 1, Q2. https://arxiv.org/html/2405.03641v1.

[7] A. A. Prikhod'ko. On ergodic flows with simple Lebesgue spectrum. *Sbornik: Mathematics* 211(4) (2020), 594–615. https://www.mathnet.ru/eng/sm8147. DOI: https://doi.org/10.1070/SM8147.

[8] B. Fayad, G. Forni, and A. Kanigowski. Lebesgue spectrum of countable multiplicity for conservative flows on the torus. *Journal of the American Mathematical Society* 34(3) (2021), 747–813. https://doi.org/10.1090/jams/970. Author manuscript examined: https://webusers.imj-prg.fr/~bassam.fayad/FFK_CILS_revised.pdf.

[9] F. Abdedou, B. Fayad, and A. Kessi. Analytic reparametrizations of translation toral flows with countable Lebesgue spectrum. *Discrete and Continuous Dynamical Systems* 43(10) (2023), 3706–3727. https://www.aimsciences.org/article/doi/10.3934/dcds.2023063.

[10] F. Abdedou, B. R. Fayad, and J.-P. Thouvenot. Inducing countable Lebesgue spectrum. *Theory of Probability and Its Applications* 70(4) (2026), 623–631; Russian original 70(4) (2025), 764–773. https://www.mathnet.ru/eng/tvp5874. Preprint: https://arxiv.org/abs/2409.00396.

[11] E. H. el Abdalaoui. *Spectral ergodic Banach problem and flat polynomials*. arXiv:1508.06439, preprint first posted 2015; the source cites version 8 (June 19, 2023). https://arxiv.org/abs/1508.06439v8. An infinite-measure claim is not used as a verified probability-space result.

[12] V. A. Rokhlin. New progress in the theory of transformations with invariant measure. *Russian Mathematical Surveys* 15(4) (1960), 1–22. Source locator: Section 7, p. 13. https://www.mathnet.ru/eng/rm6735.

[13] R. Mañé. A proof of Pesin's formula. *Ergodic Theory and Dynamical Systems* 1 (1981), 95–102. https://doi.org/10.1017/S0143385700001188. Errata: 3(1) (1983), 159–160, https://doi.org/10.1017/S0143385700001863. Imported via the source's p. 95, equation (2) locator; full proof not reread.

[14] OpenAI. *Rokhlin's multiple-mixing problem for one transformation*. September 23, 2026, Theorem 1.1. Separate public preprint cited by M279; not independently audited here. https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/Rokhlins-multiple-mixing-problem-for-one-transformation-September-23-2026/paper.pdf.